

Erdős Problem #971

EP-971

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $p(a, d)$ be the least prime congruent to $a \pmod{d}$. Does there exist a constant $c > 0$ such that, for all large d , $p(a, d) > (1 + c)\phi(d) \log d$ for $\gg \phi(d)$ many values of a ?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>